

Swiggy Food Orders

Data Analytics & Visualization

FOOD TECH

Hopper _ C2

Tableau: [[LINK1](#), [LINK2](#), [LINK3](#)]

- Kshitij Saxena
- Milan kumar
- Devanshu Prakash
- Rajat Tiwari
- Sahil Laxman Khemnagar

GitHub : [LINK](#)

Context & Problem Statement

Sector Context

India's online food delivery market exceeded ₹500 billion in 2024, with Swiggy serving 500+ cities. Restaurant owners, category managers, and city-level operations heads need data-driven insights to optimise menus, pricing, and service quality.

Problem Statement

Which cities, categories, and price segments drive the most orders and highest ratings on Swiggy — and what patterns in time and location reveal unmet demand?

Objective

Deliver a clean, analysis-ready dataset and interactive Tableau dashboards that enable business stakeholders to identify top-performing restaurants, optimise pricing tiers, and uncover temporal demand patterns across 28 Indian cities.

Data Engineering

Kaggle
swiggy_data_raw.csv
1,97,430 rows × 10 columns
Jan 2025 – Aug 2025

Load & profile raw CSV — shape, nulls, dtypes

Drop 9,871 nulls · Remove 14 duplicates · Filter 75,133 zero-rated rows · IQR outlier flag (4,180 rows >₹577)

02_cleaning.ipynb — Key transformations

```
df = df.dropna()
df = df.reset_index(drop=True)
df = df[df['Rating Count'] != 0]
df['is_price_outlier'] = ~df['Price (INR)'].between(lower, upper)
```

- ▶ state / city / location
- ▶ order_date → month, day_of_week
- ▶ price_inr → price_bucket
- ▶ rating / rating_count
- ▶ category / dish_name

Extract

Transform

Load

Warehouse

Analytics

Reporting

Dashboard

Visualization

Insights

Decision

KPI & Metrics Framework

Total Orders 91,788 Why: Measures overall platform activity and dataset coverage How: COUNT(dish_name) → 03_edu.ipynb	Total Revenue ₹2.30 Cr Why: Reflects cumulative monetary value of all listed dishes How: SUM(price_inr) → 03_edu.ipynb	Avg Price (INR) ₹250.24 Why: Benchmarks pricing strategy across cities and categories How: AVG(price_inr) → 04_statistical_analysis.ipynb
Avg Rating 4.31 / 5 Why: Tracks customer satisfaction across all restaurants How: AVG(rating) → 04_statistical_analysis.ipynb	Total Restaurants 898 Why: Scope of restaurant network across 28 cities How: COUNTD(restaurant_name) → 03_edu.ipynb	Peak Day Saturday Why: Weekends prompt for higher customer acquisition How: MAX(COUNT by day_of_week) → 05_final_load_prep.ipynb

Key Insights

Exploratory Data Analysis · 03_eda.ipynb

01

Bengaluru Dominates Order Volume

Bengaluru accounts for 10,385 orders — 1.9× more than the next city (Mumbai: 5,459), making it the single largest market on the platform.

02

Mid-Range Pricing Wins the Market

59.1% of all dishes are priced ₹150–₹400 (Mid-range). Only 11.6% fall in the Premium tier (>₹400), indicating strong price sensitivity among Swiggy users.

03

Customer Satisfaction is Consistently High

The average rating across 898 restaurants is 4.31/5.0, with a right-skewed distribution — suggesting most restaurants maintain strong quality standards.

04

Weekend Demand Peaks on Saturdays

Saturday drives the highest order volume (13,538 orders), followed by Thursday and Sunday — revealing a clear mid-week and weekend demand surge.

05

Orders are Stable Across Jan–Aug 2025

Monthly orders remain consistently between 10,876–11,882, with a slight uptick in Aug 2025 — indicating steady platform growth with no seasonal crash.

06

40% of Raw Data Flagged as Unreliable

75,133 rows had Rating Count = 0 and 9,871 rows had missing values — nearly 43% of raw data was filtered before analysis, highlighting significant data quality gaps.

Advanced Analysis

StatisticalTesting·04_statistical_analysis.ipynb·scipy library

ANOVA	Does Rating differ significantly across Cities? <code>scipy.stats.f_oneway()</code> City location has a strong statistically significant effect on restaurant ratings. Bengaluru and Chennai restaurants consistently outperform other cities — city-specific quality standards matter.
ANOVA	Does Rating differ significantly across Categories? <code>scipy.stats.f_oneway()</code> Food category influences customer satisfaction. Certain categories consistently attract higher ratings — suggesting menu specialisation drives better customer experience.
Spearman	Is there a significant correlation between Price & Rating? <code>scipy.stats.spearmanr()</code> Price and Rating are statistically linked but practically unrelated ($p = 0.029$). Higher price does NOT guarantee higher ratings — quality perception is independent of price on Swiggy.

Tableau Dashboard Walkthrough

D1

Executive Overview

KPIs: Orders · Revenue · Rating · Average Price

Charts: Top 5 Cities by Orders (Bar) · Orders by Price Bucket (Pie)

D2

Optimized Pricing Strategy | Indentify Winners

KPIs: Total Orders · Total Revenue · Avg Price · % Premium Orders

Charts: Price vs Rating Scatter · Segment Info Cards · Restaurants Table

D3

Optimized Staffing | Target Marketing

KPIs: Peak Day · Lowest Weekday · Monthly Trend

Charts: Days of Week · Month Trend · Top Categories · Segment Strategy Cards

Common Slicers (all dashboards): State · City · Category · Price Bucket · Order Date Range

Tableau URL: [Swiggy_Restaurants_Analytics](#)

Tableau URL: [Swiggy_Restaurants_Analytics2](#)

Tableau URL: [Swiggy_Restaurants_Analytics3](#)

Recommendations

R1	<i>Insight: Bengaluru generates 1.9× more orders than any other city</i> → Launch city-specific restaurant partnerships and exclusive Bengaluru deals to deepen market leadership in the top city	Impact Estimated 10–15% revenue uplift from top market
R2	<i>Insight: 59.1% of orders are Mid-range (₹150–₹400); Premium only 11.6%</i> → Introduce 'Premium Upgrade' nudges at checkout and curate a Premium restaurant collection to shift 5% of Mid-range users to Premium	Impact ~₹11.5L incremental revenue from tier migration
R3	<i>Insight: Price and Rating are statistically unrelated ($p = 0.029$)</i> → Market budget restaurants with high ratings more aggressively — value-for-money positioning is a proven driver of orders independent of price	Impact Higher conversion among price-sensitive users
R4	<i>Insight: Saturday & Thursday drive peak demand; Monday is lowest</i> → Launch weekday promotions (Monday–Tuesday discounts) and scale delivery capacity on Saturdays to reduce drop-offs during peak hours	Impact 15–20% reduction in order cancellations on peak days
R5	<i>Insight: 43% of raw data had quality issues (nulls + zero ratings)</i> → Implement mandatory restaurant data completion checks before listing — enforce minimum rating count thresholds for visibility on the platform	Impact Cleaner data → more reliable recommendations

Impact & Value

The "So What?" — Why stakeholders must act on these findings

Total Revenue
Analysed

Raw Data Cleaned
Before Analysis

Platform Avg
Rating

Cities &
States Covered

Cost of Inaction

Without data-driven decisions, Swiggy risks losing high-value markets like Bengaluru to competitors who invest in localised restaurant partnerships and targeted pricing.

Revenue Opportunity

Shifting 5% of Mid-range users to Premium tier across 91,788 orders could unlock ~₹11.5L in incremental revenue — achievable through checkout nudges alone.

Operational Efficiency

Scaling delivery capacity on peak days (Saturday +13.5K orders) and reducing Monday idle time can cut operational costs by an estimated 12–18% through demand-matched staffing.

Data Quality ROI

Enforcing minimum data standards at listing could recover 43% of currently unusable data — directly improving recommendation accuracy and customer experience scores.

Limitations & Next Steps

Limitations

Dataset is synthetic — findings may not reflect real Swiggy platform behaviour
43% of raw data dropped — analysis based on 91,788 of 1,97,430 original rows
No customer-level data — cannot analyse repeat orders or churn behaviour
No delivery time or complaint data — service quality analysis is incomplete
Order Date range limited to Jan–Aug 2025 — no seasonal trend beyond 8 months

Next Steps

01	Build a predictive model for restaurant rating using price, category, and city features (sklearn)
02	Integrate real-time Swiggy API data for live dashboard updates in Tableau
03	Extend dataset to full year (Jan–Dec) for complete seasonal trend analysis
04	Add customer segmentation (RFM model) once order-level customer data is available
05	Deploy ETL pipeline as a scheduled job (Apache Airflow) for automated weekly refresh

Team & Contribution

01_extraction · 02_cleaning

RAJAT TIWARI

03_eda

KSHITIJ SAXENA

04_statistical_analysis

DEVANSHU PRAKASH

SAHIL LAXMAN
KHEMNAR

KSHITIJ SAXENA

MILAN KUMAR